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CIS 4360 Section 1
Lab 5
10/19/2026

```
tuk97813@cloudshell:~ (cloud-native-507602)$
tuk97813@cloudshell:~ (cloud-native-507602)$ gcloud config get-value compute/zone
Your active configuration is: [cloudshell-17152]
(unset)
tuk97813@cloudshell:~ (cloud-native-507602)$ gcloud config get-value project
Your active configuration is: [cloudshell-17152]
cloud-native-507602
tuk97813@cloudshell:~ (cloud-native-507602)$ gcloud config set compute/zone us-east1-b
Updated property [compute/zone].
tuk97813@cloudshell:~ (cloud-native-507602)$ gcloud config get-value compute/zone
Your active configuration is: [cloudshell-17152]
us-east1-b
tuk97813@cloudshell:~ (cloud-native-507602)$
```

Screenshot shows verifying the project name and zone. The zone was not set, so we run the command to set it to us-east1-b again.

```
tuk97813@cloudshell:~ (cloud-native-507602)$ gcloud container clusters list
tuk97813@cloudshell:~ (cloud-native-507602)$ gcloud container clusters create cluzter --num-nodes=3
 --machine-type=e2-medium
Note: Your Pod address range (`--cluster-ipv4-cidr`) can accommodate at most 1008 node(s).
Creating cluster cluzter in us-east1-b... Cluster is being health-checked (Kubernetes Control Plan
e is healthy)...done.
Created [https://container.googleapis.com/v1/projects/cloud-native-507602/zones/us-east1-b/clusters
/cluzter].
To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workloa
d_/gcloud/us-east1-b/cluzter?project=cloud-native-507602
kubeconfig entry generated for cluzter.
NAME: cluzter
LOCATION: us-east1-b
MASTER_VERSION: 1.36.4-gke.1495002
MASTER_IP: 104.196.52.94
MACHINE_TYPE: e2-medium
NODE_VERSION: 1.36.4-gke.1495002
NUM_NODES: 3
STATUS: RUNNING
STACK_TYPE: IPV4
```

```
tuk97813@cloudshell:~ (cloud-native-507602)$ gcloud container clusters get-credentials cluzter --zo
ne=us-east1-b
Fetching cluster endpoint and auth data.
kubeconfig entry generated for cluzter.
tuk97813@cloudshell:~ (cloud-native-507602)$
```

Screenshot shows us checking if any clusters exist (they don't), and creating and authenticating a new cluster called "cluster" with 3 nodes.

```
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ nano nginx-deployment.yaml
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
        resources:
          requests:
            cpu: "200m"
          limits:
            cpu: "500m"
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$
```

Screenshot shows adding a deployment file, along with the “resources” field that contains the requests and limits for cpu usage (which if the utilization rate goes over 50% of what’s specified, the HPA will kick in and auto scale for us.

```
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

```
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ kubectl expose deployment nginx-deployment --por
=80 --target-port=80
service/nginx-deployment exposed
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$
```

And going ahead and applying and exposing it to our deployment.

```
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ kubectl get nodes
NAME                                          STATUS   ROLES    AGE   VERSION
gke-cluzter-default-pool-7f096b0d-067b   Ready    <none>   21m   v1.36.4-gke.1495002
gke-cluzter-default-pool-7f096b0d-qwn7   Ready    <none>   21m   v1.36.4-gke.1495002
gke-cluzter-default-pool-7f096b0d-rl5t   Ready    <none>   21m   v1.36.4-gke.1495002
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   1/1     1            1           4m47s
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-86c65d7c74-x5v5c   1/1     Running   0          4m55s
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$
```

Showing the set up nodes, deployments, and pods. We see we have 3 nodes along with our pod.

```
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ kubectl describe deployment nginx-deployment
Name:                   nginx-deployment
Namespace:              default
CreationTimestamp:      Thu, 22 Oct 2026 04:49:08 +0000
Labels:                 app=nginx
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=nginx
Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=nginx
  Containers:
   nginx:
    Image:      nginx:latest
    Port:       80/TCP
    Host Port:  0/TCP
    Limits:
      cpu:  500m
    Requests:
      cpu:         200m
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   nginx-deployment-86c65d7c74 (1/1 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  9m45s  deployment-controller  Scaled up replica set nginx-deployment-8
6c65d7c74 from 0 to 1
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$
```

Here we see the details of our full setup. We can see we have our deployment set up. We see the Limits: and Requests: have been set. We also see we have a ReplicaSet set up which is where our pods live (assuming more will be created as the HPA scales up).

```
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ kubectl run temp-client --rm -it --image=alpine --
- sh
All commands and output from this session will be recorded in container logs, including credentials
 and sensitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
/ # wget -qO- http://nginx-deployment
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit
Session ended, resume using 'kubectl attach temp-client -c temp-client -n default -i -t' command
pod "temp-client" deleted from default namespace
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$
```

In this screenshot we made a temporary pod, and tested connectivity to the existing running pod. This shows the pods can communicate to each other within the cluster.

```
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ kubectl delete -f nginx-deployment.yaml
deployment.apps "nginx-deployment" deleted from default namespace
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ kubectl delete service nginx-deployment
service "nginx-deployment" deleted from default namespace
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$ gcloud container clusters delete cluzter --zone=u
s-east1-b
The following clusters will be deleted.
 - [cluzter] in [us-east1-b]

Do you want to continue (Y/n)?  y

Deleting cluster cluzter...done.
Deleted [https://container.googleapis.com/v1/projects/cloud-native-507602/zones/us-east1-b/clusters
/cluzter].
tuk97813@cloudshell:~/lab5 (cloud-native-507602)$
```

In this screenshot we show the deletion and cleanup. We removed the deployment, deleted the serviced, and deleted the cluster.